ECG Trust Layer

Which heartbeats of a digitized paper ECG can you still verify against the image? · Built for UnivaBio 2026 · research prototype, not a medical device

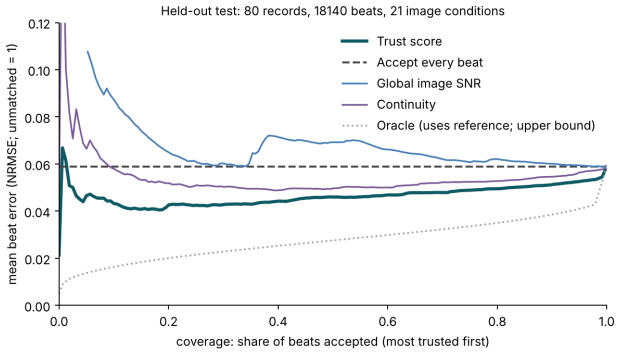
What it does

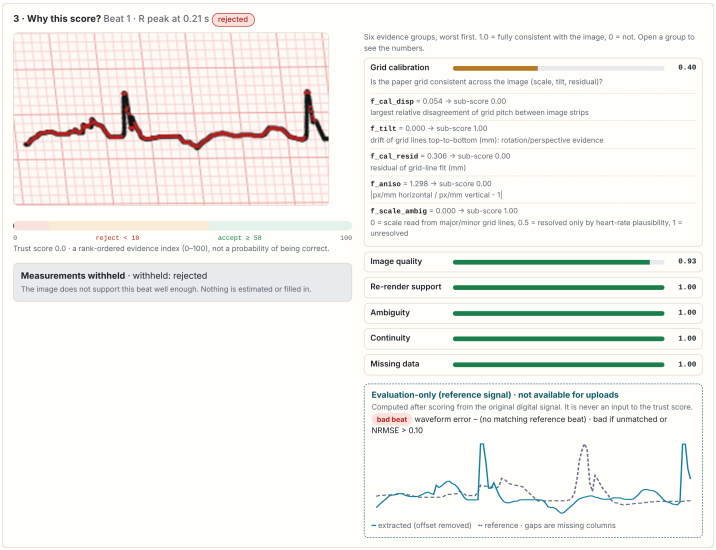
Digitizers turn a picture of a paper ECG into a signal, and downstream numbers trust it blindly. ECG Trust Layer gives **every heartbeat a trust score computed from the image alone** (no reference signal) and reports R amplitude, RR interval and heart rate **only for verifiable beats**. Unreadable columns stay explicit gaps; unreadable images return “unable”. A local web app shows the image, the extracted trace, per-beat decisions, an evidence inspector, and a real “Reproduce the Evaluation” button.

Result: held-out test, 80 PTB-XL records, 18,140 beats, 21 image conditions

Quantity	AURC (mean beat error over coverage; lower is better)
Accept every beat	0.0589
Best simple baseline (continuity)	0.0547
Trust score (learned)	0.0468
H1: trust beats accept-all	SUPPORTED. AURC lower by 0.0121 (-20%); 95% CI of the improvement [0.0059, 0.0175]
H2: trust beats best baseline	SUPPORTED NARROWLY. Lower by 0.0079; CI [0.0015, 0.0139]. Not supported on validation.

At the app's **post-hoc** operating point, **70% of beats are accepted with a 1.6% bad-beat rate (4.1% overall)**.





Left: risk-coverage on the test split. Right: the evidence inspector for a rejected beat (tilted image).

What did not work (reported honestly)

- Our **pre-registered operating-point rule was vacuous** (it accepted every beat); reported unchanged, with one clearly labelled post-hoc replacement.
- No help for severe blur; 15% of accepted beats are still bad under severe thick-trace.
- Withholding costs: 30% of clean beats withheld; 23% of analyzable images have no reportable heart rate.
- Global image SNR is worse than accepting everything.

Limits you must know

All images are synthetic renderings of PTB-XL lead II, not photographs of real paper ECGs; real-world performance is unknown. No clinical validation; not for diagnosis. The score is a rank-ordered evidence index, not a probability. Single lead, 10 s, standard grid, speed and gain assumed.

Reproduce

`python scripts/run_evaluation.py --preset quick` (about 30 s) · `python app/server.py` · 66 automated tests · code MIT; data PTB-XL v1.0.3, CC BY 4.0 (Wagner et al., Sci Data 2020). The demo narration is synthetic speech (Piper TTS). github.com/Abdullah49645/ecg-trust-layer